INTERNNOVA • WEEK 6 ASSIGNMENT

FINAL DATA ANALYTICS PROJECT REPORT

Task-wise Documentation | Git • Python • EDA • Visualization • Power BI

Student Name	Nakul Mehra
Internship	InternNova – Data Analytics Internship
Dataset	Sample Superstore — cleaned Excel dataset
GitHub Repository	Nakulmehra19/week6-final-data-analytics-project

Prepared according to the Week 6 assignment tasks and marking scheme

Report Contents

- Task 1: Git & GitHub Basics (10 Marks)
- Task 2: Git Workflow Practice (10 Marks)
- Task 3: Dataset Selection & Data Preparation (10 Marks)
- Task 4: Exploratory Data Analysis (EDA) (15 Marks)
- Task 5: Data Visualization (15 Marks)
- Task 6: Power BI Dashboard (20 Marks)
- Task 7: Business Insights & Recommendations (10 Marks)
- Task 8: GitHub Documentation & Project Presentation (10 Marks)
- Conclusion
- Appendix: Final Submission Checklist

Project snapshot

Dataset: Sample Superstore | Working dataset: 9,994 rows × 21 columns | Tools: Python, Pandas, NumPy, Matplotlib, Seaborn, Excel, Power BI, Git and GitHub.

Task 1: Git & GitHub Basics (10 Marks)

Objective: Set up Git and GitHub and use a remote repository to store the final data analytics project.

Work completed

- Git and GitHub were used as the version-control and project-hosting workflow for the capstone.
- A new GitHub repository was created for the Week 6 final project.
- The project was worked on locally in the folder week6-final-data-analytics-project.
- Project scripts, notebooks, visualizations, Power BI dashboard and README documentation were organized for repository hosting.
- Changes were committed and pushed to the remote GitHub repository.

Requirement	Project evidence / status
Install Git	Git workflow was used during project development; keep installation screenshot if required by the evaluator.
Create GitHub account	GitHub account/repository is in use for this project.
Create repository	week6-final-data-analytics-project
Clone repository	Local project folder was used; include a terminal screenshot of git clone if separately required.
Add project files	Project scripts, notebooks, visualizations, dashboard and documentation were added.
Commit changes	Multiple project commits were made during development.
Push to GitHub	Latest project files and README/presentation were uploaded to GitHub.

Repository
https://github.com/Nakulmehra19/week6-final-data-analytics-project

Suggested evidence screenshots

- GitHub repository home page showing repository name and files.
- Local project folder showing project files.
- Git terminal showing a successful commit and push.

Figure 2. Repository files and project contents visible on GitHub.

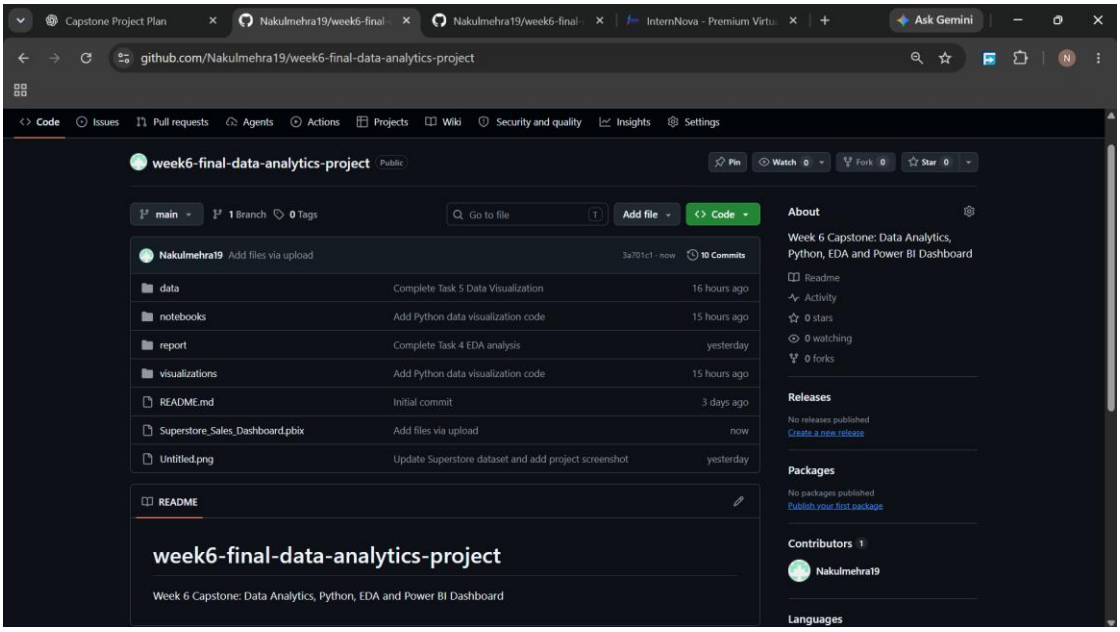
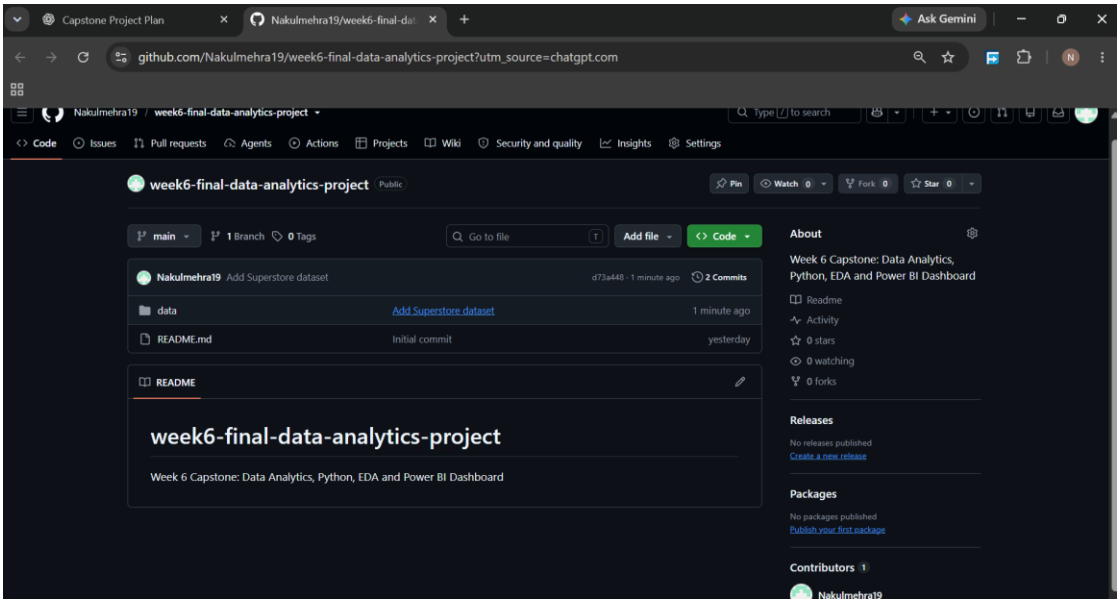


Figure 1. GitHub repository created for the Week 6 project.



Task 1 Screenshots — GitHub Repository

Task 2: Git Workflow Practice (10 Marks)

Objective: Practice the basic Git workflow and create at least three meaningful commits while developing the final project.

Git command	Purpose in the workflow
git clone <repository-url>	Copies the remote repository to the local computer.
git add .	Stages changed and new files for the next commit.
git commit -m "message"	Records a named snapshot of staged changes.
git push	Uploads local commits to GitHub.
git pull	Fetches and integrates remote changes into the local branch.

Project workflow summary

1. Create or clone the project repository.
2. Add the data preparation and analysis files.
3. Commit and push the analysis/EDA work.
4. Add the visualization script and generated charts, then commit and push.
5. Add or update the Power BI dashboard and project documentation/presentation, then commit and push.

The project history previously showed commits for EDA and visualization work (including commit identifiers 22b3dbd and fd574f2). Additional repository changes included the dashboard, README documentation and presentation. Use the GitHub Commits page as the final evidence of the complete commit history.

Evidence note
Only show commands that you actually ran. If the evaluator requires proof of git pull, take a screenshot of the terminal after running it successfully.

Figure 4. Local project development workflow.

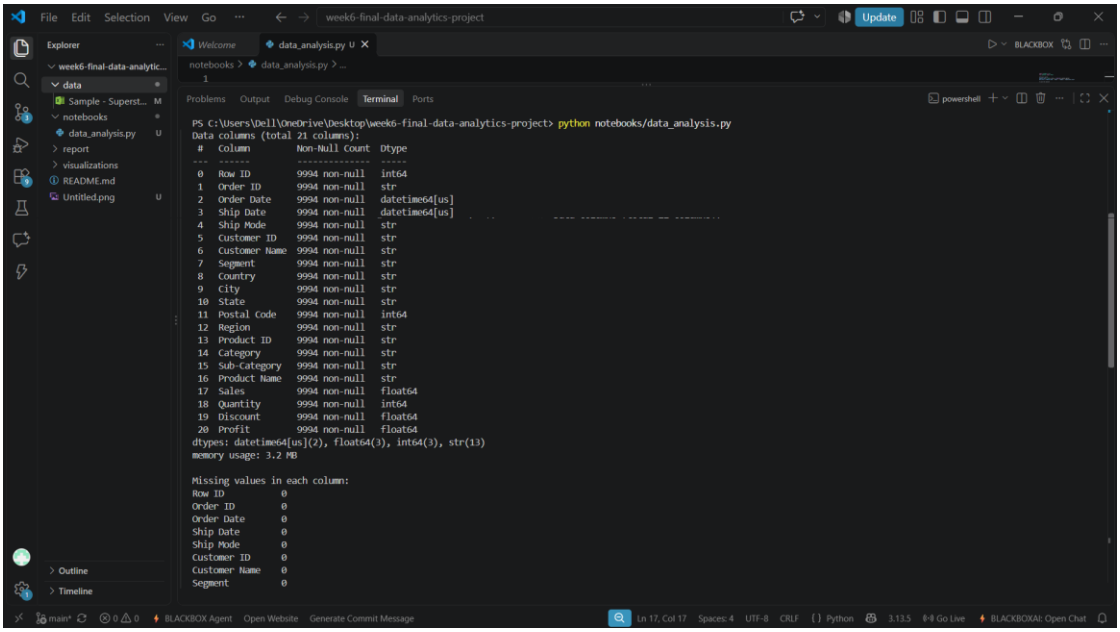
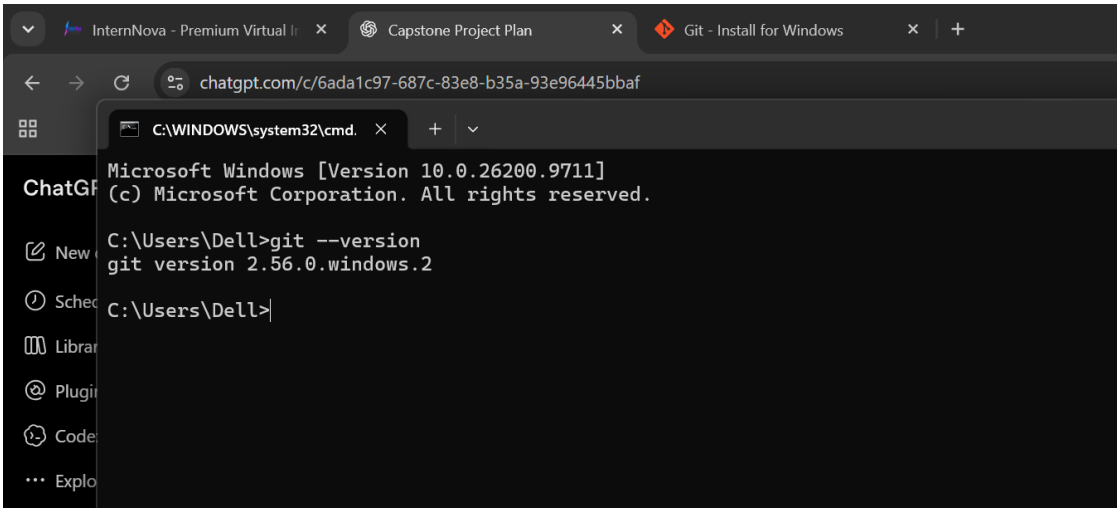


Figure 3. Terminal evidence from project setup and Git workflow.



Task 2 Screenshots — Git Workflow

Task 3: Dataset Selection & Data Preparation (10 Marks)

Selected domain: Retail sales / e-commerce analytics.

Dataset property	Details
Dataset name	Sample Superstore
Working file	Sample - Superstore-Cleaned.xlsx
Dataset size	9,994 rows and 21 columns
Purpose	Analyze sales, profit, product categories, regions, customer segments and trends.
Main field groups	Order dates, shipping dates, order/customer identifiers, geography, product/category, sales, quantity, discount and profit.

Preparation workflow

6. Load the cleaned Excel dataset into the Python/analysis workflow.

7. Inspect the number of rows and columns, field names and data types.
8. Review missing values and duplicate records.
9. Check numeric and categorical columns for consistency.
10. Ensure date fields use a suitable date type for time-based analysis.
11. Prepare the working data for EDA, chart generation and Power BI reporting.

The project uses a cleaned working workbook. This report describes the preparation checks reflected in the project documentation and does not claim that every missing value or duplicate was removed unless a specific validation result was recorded.

Evidence to include

- Screenshot of the dataset opened in Excel or loaded in Python.
- Screenshot of dataset shape and data types.
- Screenshot/output of missing-value and duplicate checks, if available.

Figure 6. Local project files used for dataset preparation.

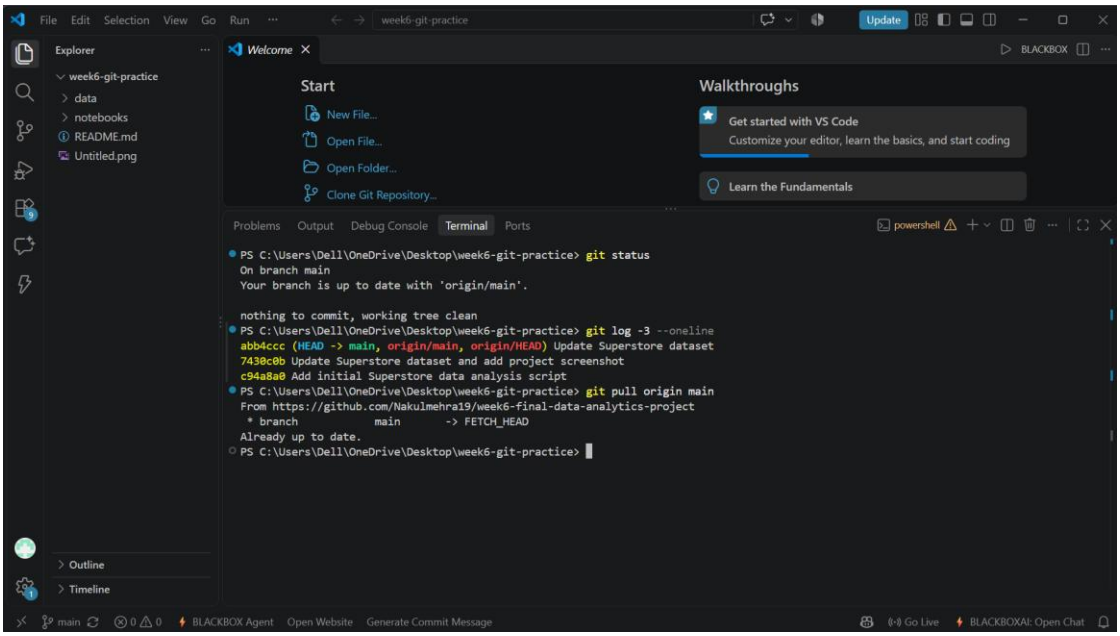
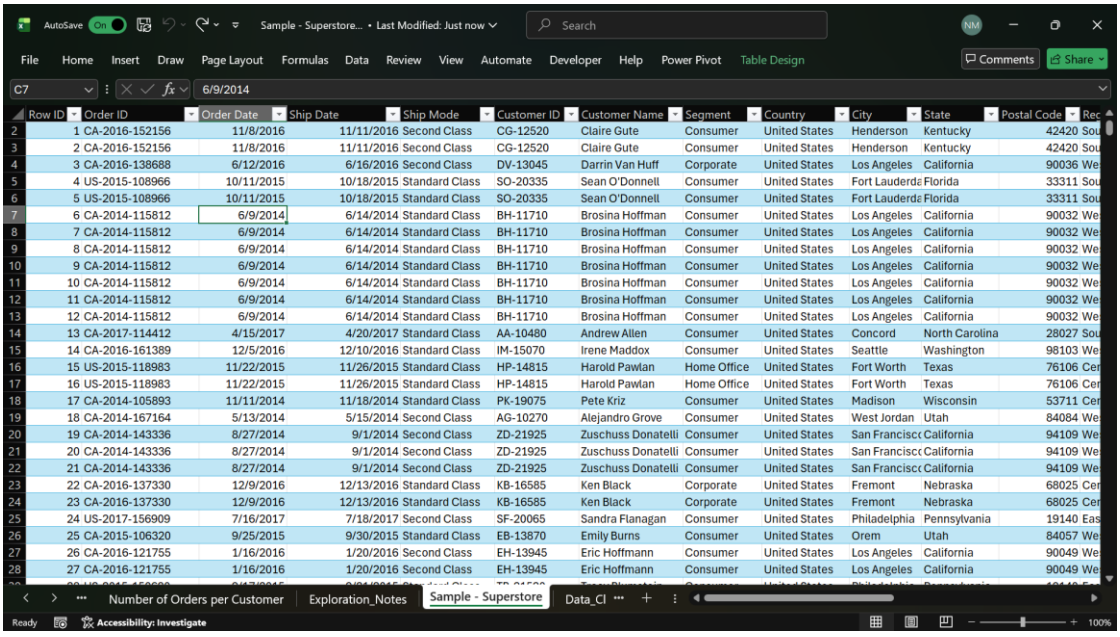


Figure 5. Sample Superstore dataset opened for inspection.



Task 3 Screenshots — Dataset Preparation

Task 4: Exploratory Data Analysis (EDA) (15 Marks)

EDA was used to inspect the data and explore patterns relevant to business performance.

EDA component	What it helps evaluate
Descriptive statistics	Central tendency, spread and ranges of numeric fields.
Data inspection	Dataset structure, column names, row count and data types.
Data quality checks	Potential missing values, duplicates and inconsistent fields.
Correlation analysis	Relationships among selected numerical variables; correlation does not imply causation.
Pattern and trend analysis	Sales comparisons by category, region, time and customer segment.
Outlier analysis	Potential extreme values in sales, profit, quantity and discount distributions.

Key EDA focus areas

- Sales and profit distributions.
- Category-wise and region-wise sales.
- Monthly sales patterns.
- Customer-segment contribution.
- Relationships among numerical fields.
- Potential outliers shown by boxplots.

The EDA outputs support the project's charts and Power BI dashboard. Specific statistical values and exact monthly peaks should be copied from the actual script output or chart rather than estimated.

Evidence to include

- Screenshot of Python/Jupyter EDA output.
- Screenshot of descriptive statistics or correlation matrix.
- Screenshot of missing-value/duplicate checks if recorded.

Figure 8. Additional analysis output from the project workflow.

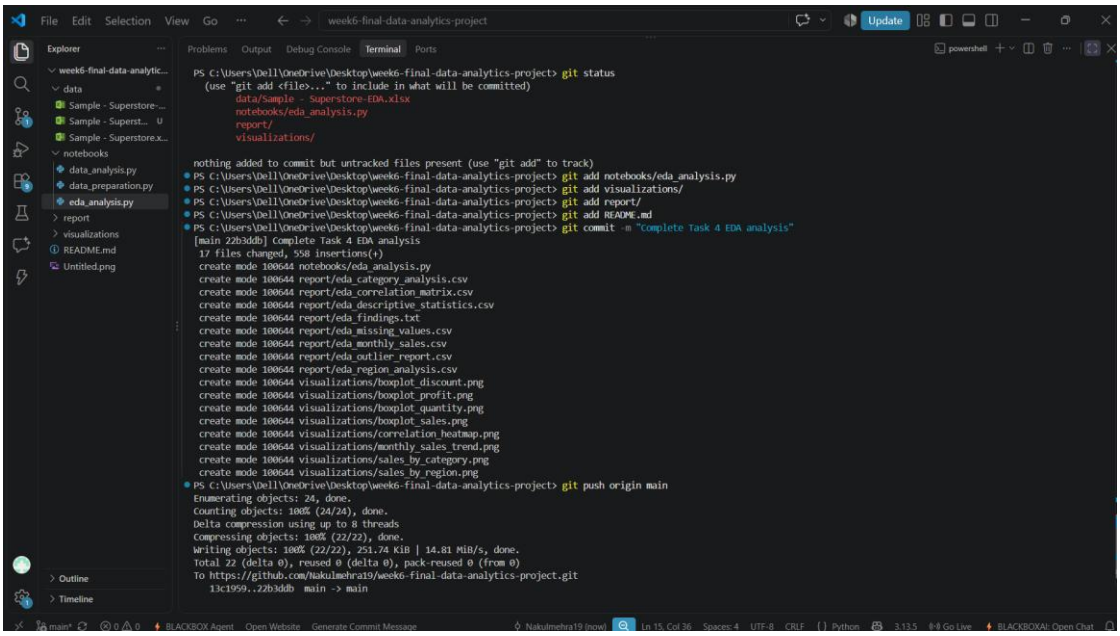
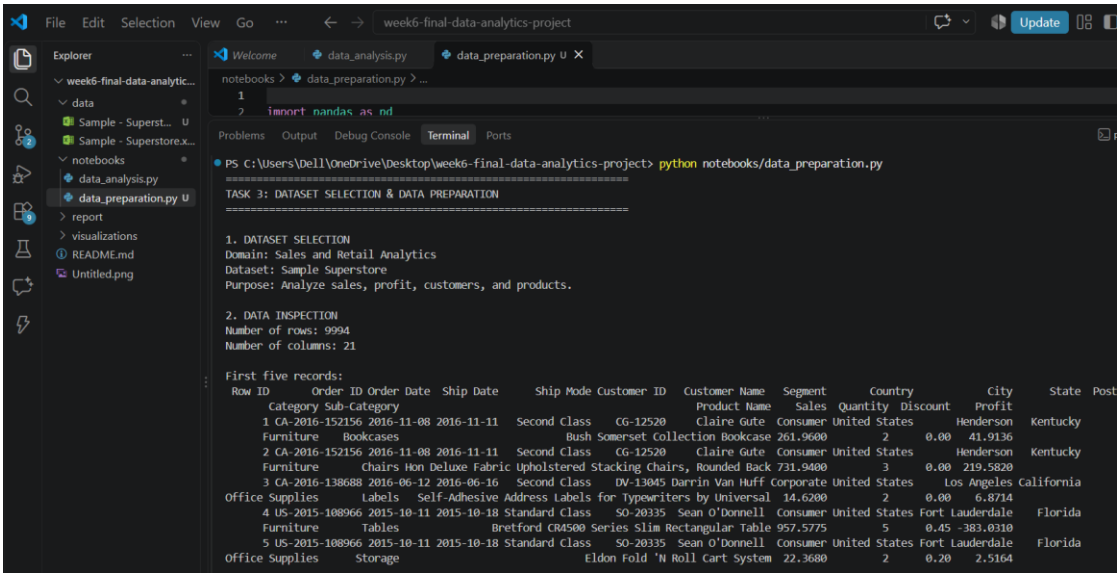


Figure 7. Python analysis output during exploratory data analysis.



Task 4 Screenshots — EDA

Task 5: Data Visualization (15 Marks)

Eight Python visualization outputs were generated for the project, exceeding the assignment minimum of five charts.

No.	Visualization file	Purpose
1	sales_by_category.png	Compare sales across product categories.
2	sales_by_region.png	Compare sales across geographic regions.
3	monthly_sales_trend.png	Show how sales vary over time.
4	correlation_heatmap.png	Explore relationships among numerical variables.
5	boxplot_sales.png	Inspect spread and potential outliers in sales.
6	boxplot_profit.png	Inspect spread and potential outliers in profit.
7	boxplot_quantity.png	Inspect quantity distribution.
8	boxplot_discount.png	Inspect discount distribution and possible extremes.

The charts are stored in the visualizations folder. The category and region charts support comparisons, the monthly trend supports time-based planning, the heatmap summarizes numerical relationships, and the boxplots highlight distributions and possible extreme observations.

Report screenshot placement
Insert the actual PNG chart images here if the instructor specifically requires the visuals embedded in the report. The image files should be taken from the project's visualizations/ folder; they were not attached to this chat as image files.

Figure 16. Discount distribution.

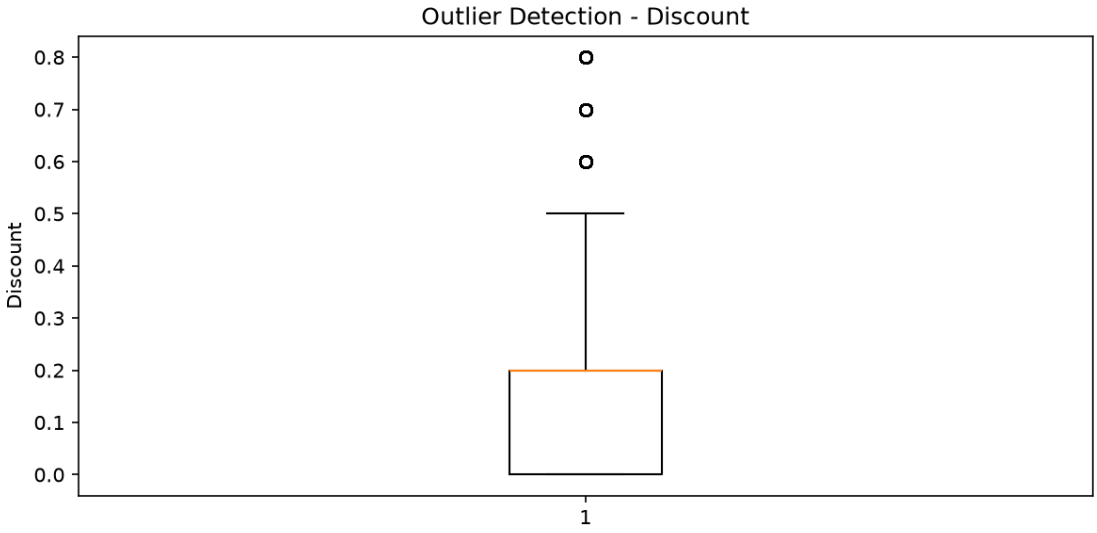


Figure 15. Quantity distribution.

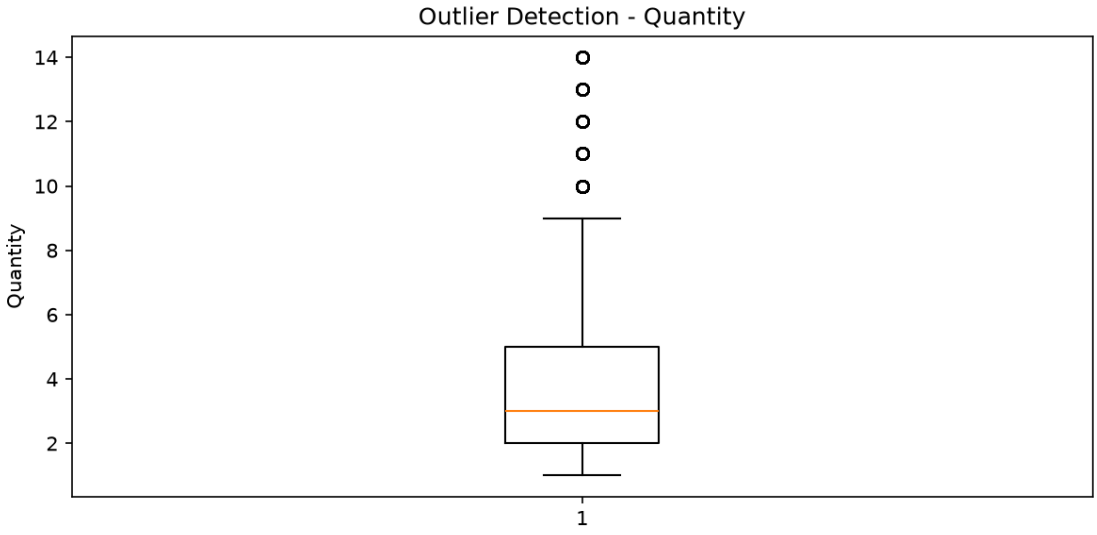


Figure 14. Profit distribution and potential outliers.

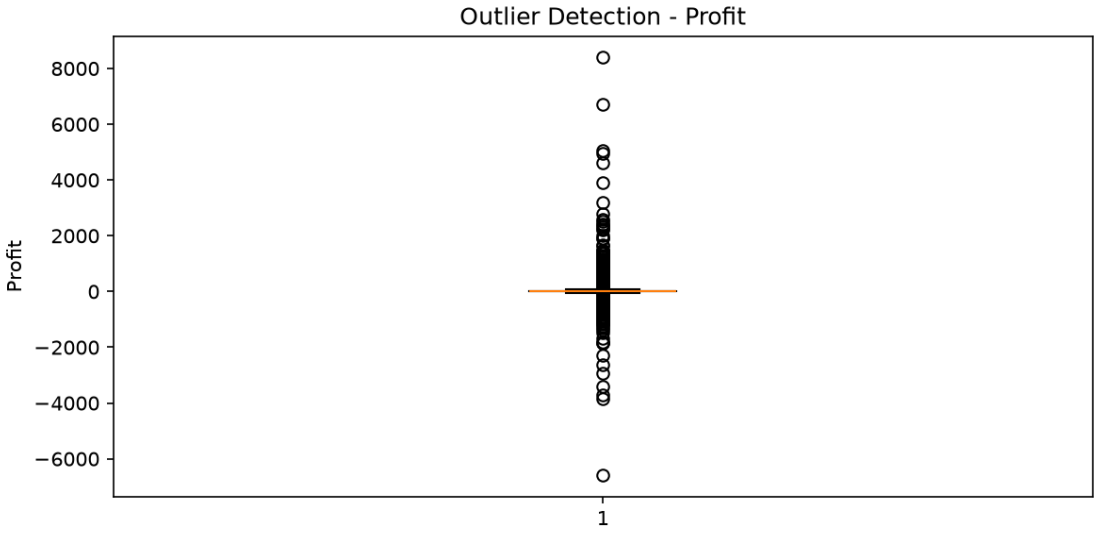


Figure 13. Sales distribution and potential outliers.

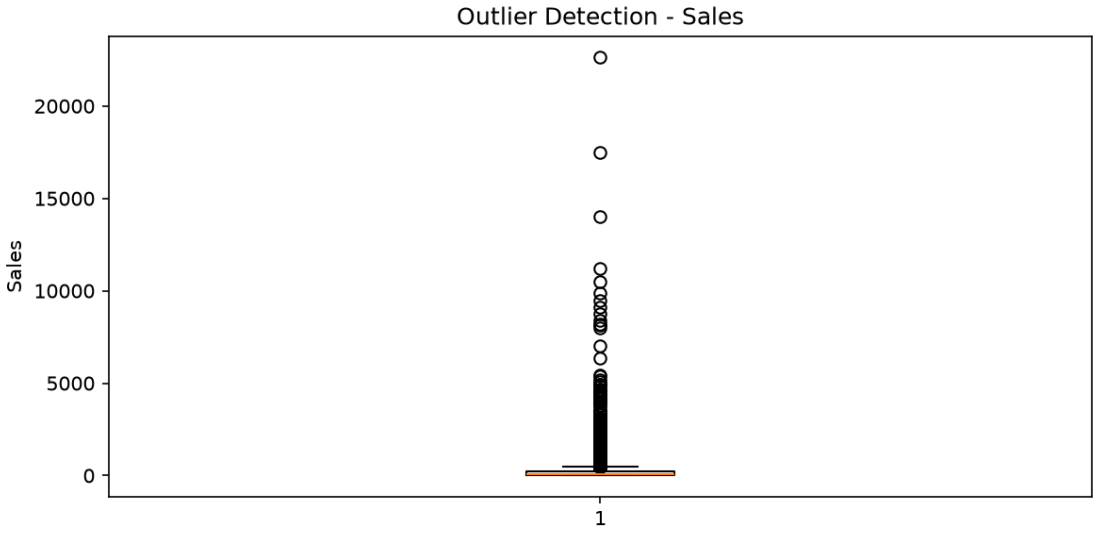


Figure 12. Correlation heatmap.

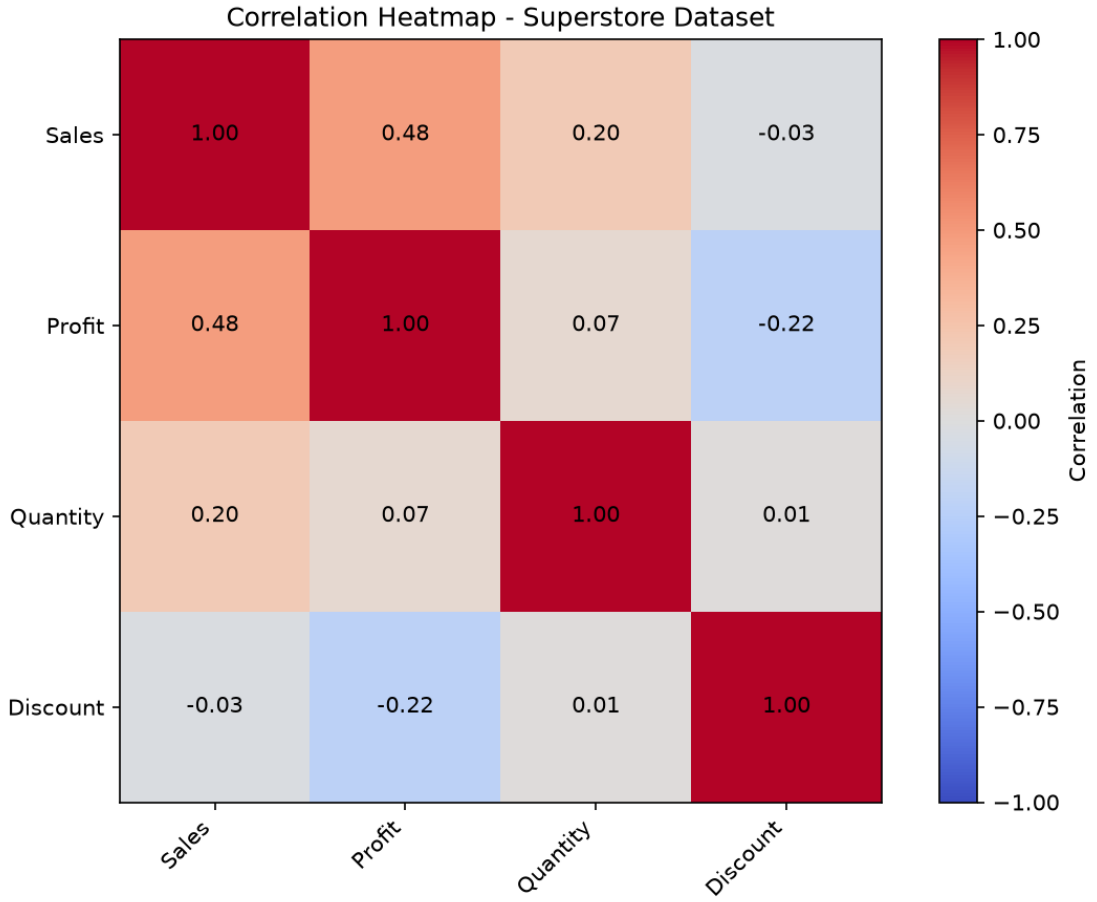


Figure 11. Monthly sales trend.

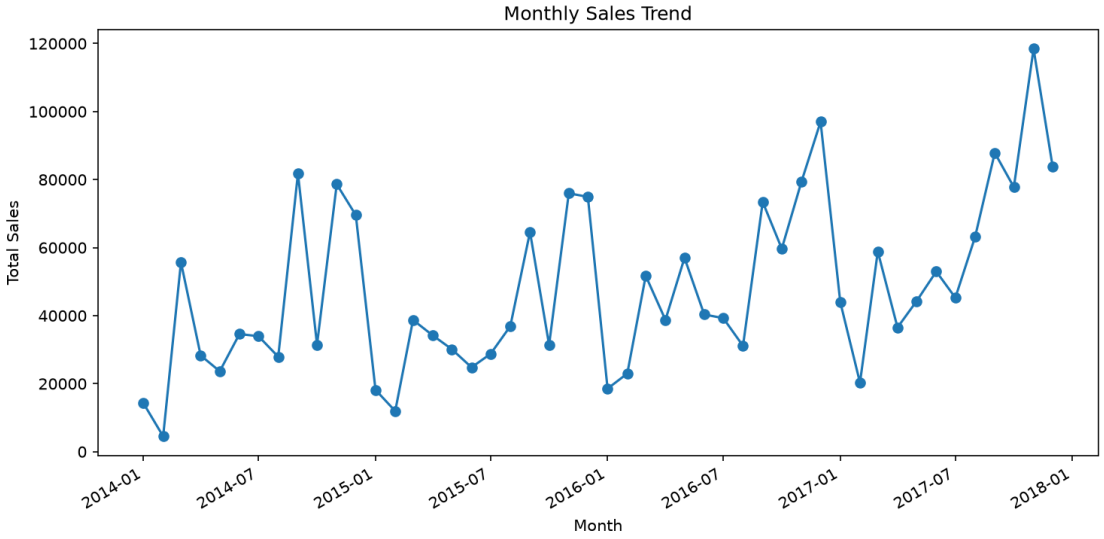


Figure 10. Sales by region.

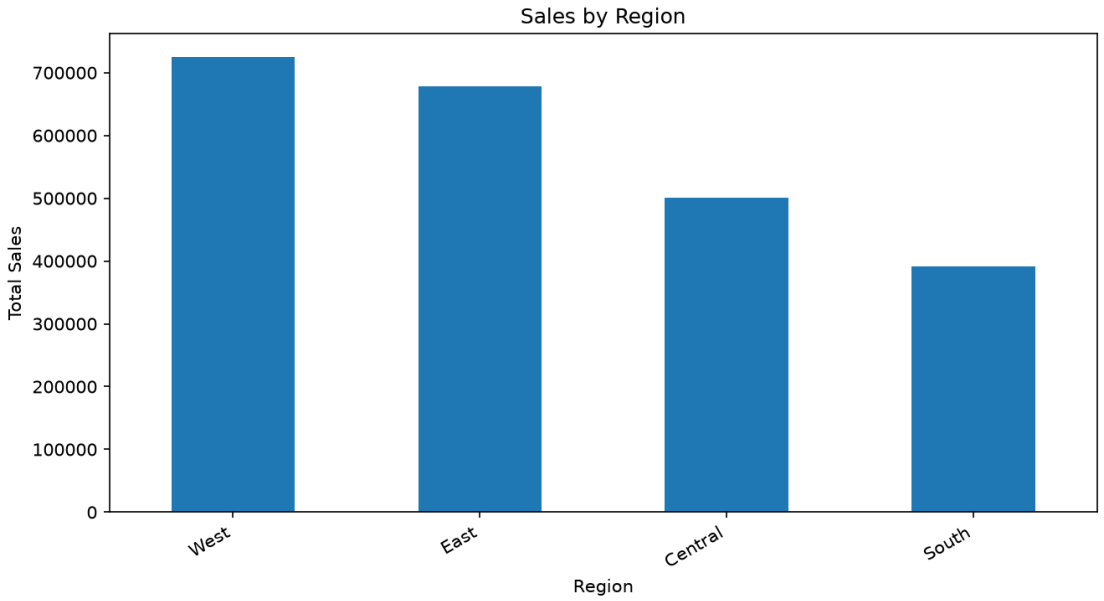
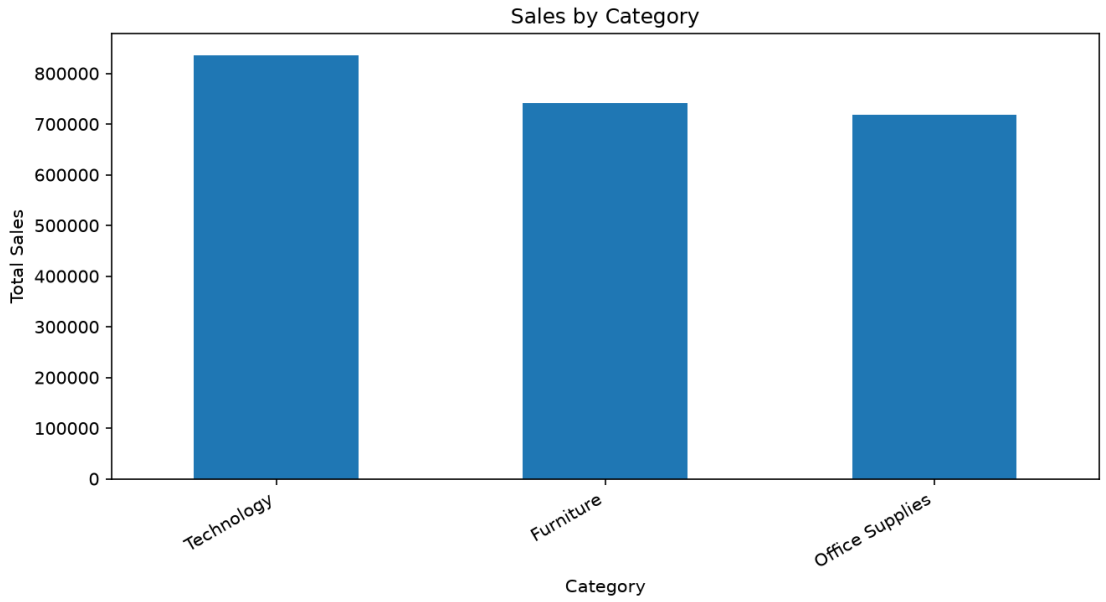


Figure 9. Sales by category.



Task 5 Screenshots — Python Visualizations

Task 6: Power BI Dashboard (20 Marks)

An interactive Power BI dashboard was created using the Sample Superstore dataset.

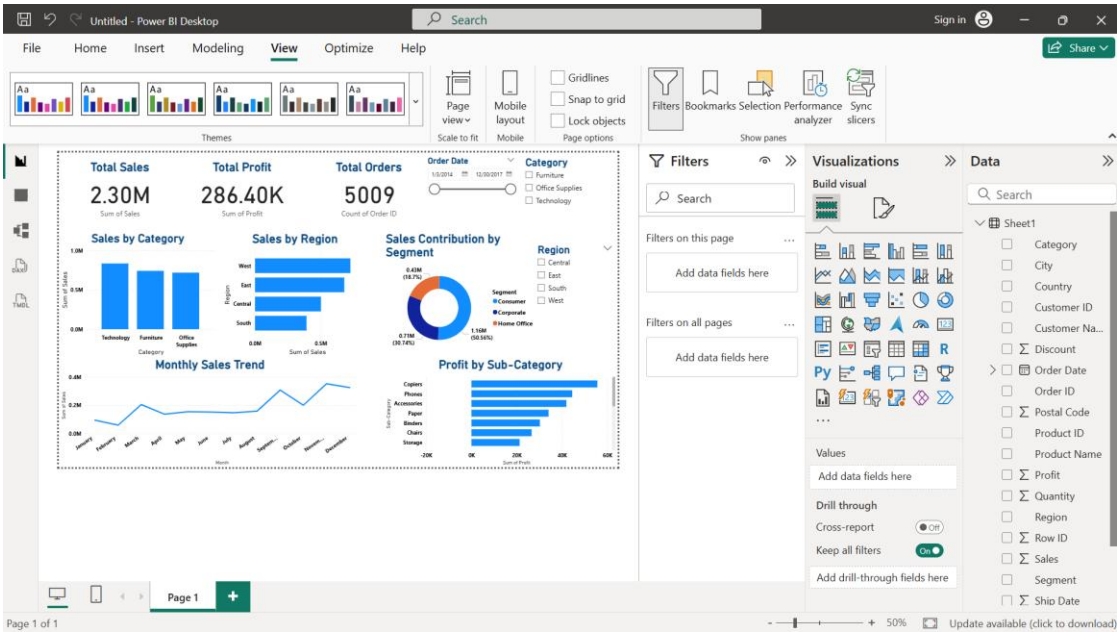
Assignment requirement	Project implementation
At least 3 KPI cards	Total Sales, Total Profit and Order Count.
At least 4 visualizations	Sales by Category; Sales by Region; Monthly Sales Trend; Profit by Sub-Category; Sales Contribution by Segment.
Filters / slicers	Order Date, Category and Region.
Meaningful title	Superstore Sales & Profit Analysis Dashboard.
Layout and organization	KPI cards at the top with comparison charts and slicers for interactive filtering.
Dashboard file	Superstore_Sales_Dashboard.pbix.

Dashboard values from the shared view

Metric / chart	Observed value or finding
Total Sales	Approximately 2.30M.
Total Profit	Approximately 286.40K.
Order Count	Displayed as 5,009; verify whether the card is a distinct Order ID count or another aggregation.
Sales by Category	Technology highest; Furniture next; Office Supplies lower.
Sales by Region	West highest; South lowest.
Segment contribution	Consumer 1.16M (50.56%); Corporate 0.71M (30.74%); Home Office 0.43M (18.70%).
Profit by Sub-Category	Copiers appeared among stronger contributors in the reviewed chart; confirm exact ranking from the final dashboard.

Before submission
Open the PBIX file, reset all slicers, verify the Order Count aggregation, and capture a clear screenshot of the full dashboard. Add that screenshot to the report if required.

Figure 17. Power BI dashboard with KPIs, charts and slicers.



Task 6 Screenshot — Power BI Dashboard

Task 7: Business Insights & Recommendations (10 Marks)

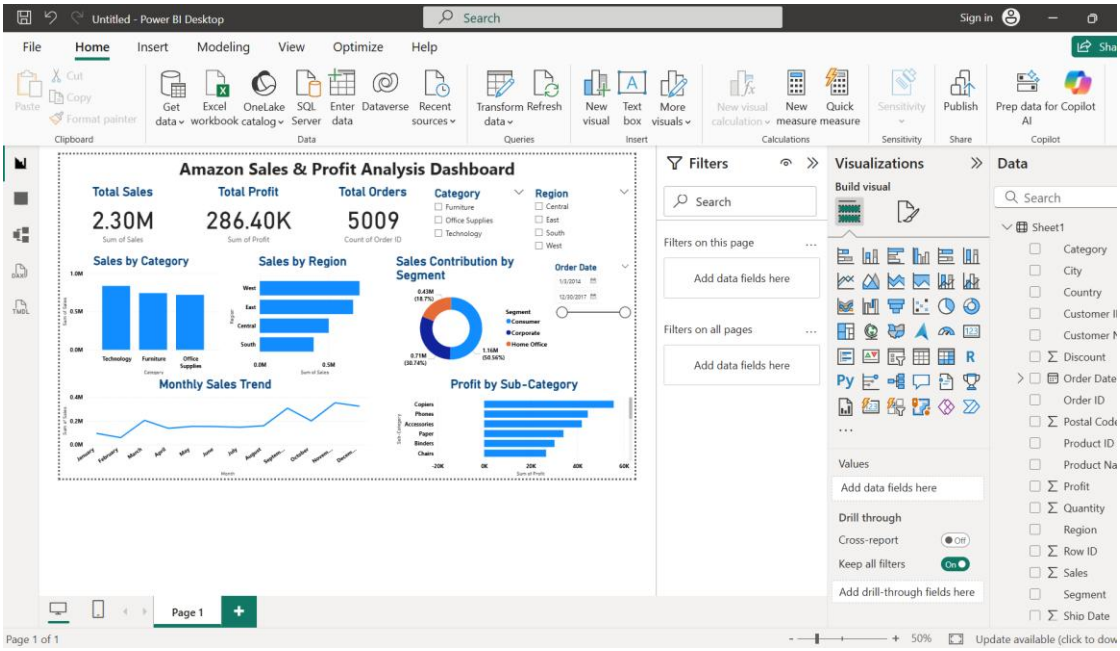
7.1 Business insights

- **Category performance:** Technology has the highest sales among the three main categories; Furniture follows, while Office Supplies has lower sales in the reviewed dashboard.
- **Regional performance:** West has the highest sales, while South has the lowest among the regions in the reviewed dashboard.
- **Customer segments:** Consumer contributes 50.56% of displayed sales, Corporate 30.74%, and Home Office 18.70%.
- **Sales over time:** Monthly sales vary; specific peak and low months should be identified directly from the monthly trend chart.
- **Sub-category profitability:** Copiers appeared among stronger profit contributors in the reviewed chart; confirm exact values and ranking.
- **Revenue versus profit:** Sales and profit are different metrics, so high sales should not automatically be interpreted as high profitability.

7.2 Data-driven recommendations

12. **Investigate the South region:** Review demand, customer feedback, product availability and marketing activity before choosing corrective actions.
13. **Plan inventory using category demand:** Maintain suitable inventory for strong-selling Technology products and explore ways to improve sales in other categories.
14. **Use monthly patterns for planning:** Use observed trends to plan inventory, staffing and promotions for high-demand periods.
15. **Monitor profitability:** Review profit by sub-category alongside discounts and costs; investigate low-margin areas.
16. **Tailor campaigns by segment:** Use segment filters and observed sales contribution to guide customer-specific marketing tests.

Figure 18. Supporting project screenshot for final analysis and recommendations.



Task 7 Screenshot — Business Insights

Task 8: GitHub Documentation & Project Presentation (10 Marks)

The GitHub README documents the project and the PowerPoint presentation summarizes the workflow and findings. The repository was reviewed through screenshots shared during the project.

Required documentation item	Status observed in README screenshots
Project Title	Present: Superstore Sales & Profit Analysis.
Project Overview	Present.
Problem Statement	Present.
Dataset Description	Present; Sample Superstore, cleaned workbook, 9,994 rows.
Tools Used	Present.
Data Cleaning Process	Present.
EDA	Present.
Visualizations	Present; visualization file names listed.
Power BI Dashboard	Present; dashboard file and components described.
Key Insights	Present.
Business Recommendations	Present.
Conclusion	Present.
Project Structure / How to Use	Also present in screenshots.
Presentation	User confirmed PowerPoint uploaded to GitHub.

Presentation outline

- Project title and overview.
- Problem statement and objectives.
- Dataset and tools.
- Data preparation and EDA.
- Python visualizations.
- Power BI dashboard and KPIs.
- Five or more business insights.
- Recommendations, limitations and conclusion.

Suggested evidence: a screenshot of the GitHub repository file list, README ending, presentation file in the repository, and the presentation cover/slide view.

Figure 20. README business recommendations, project structure and conclusion.

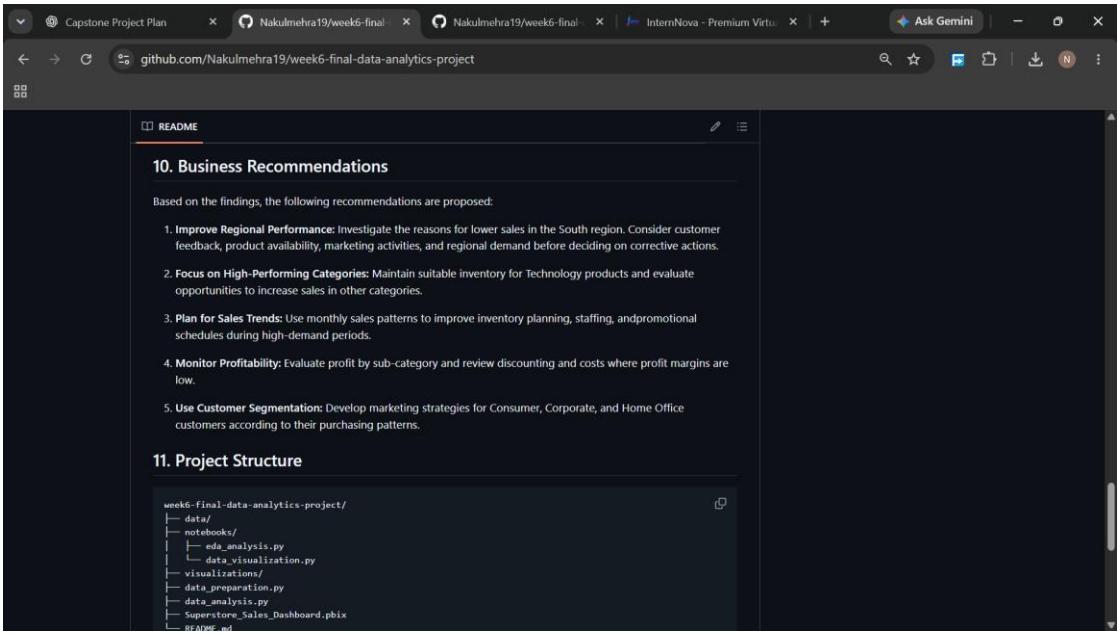
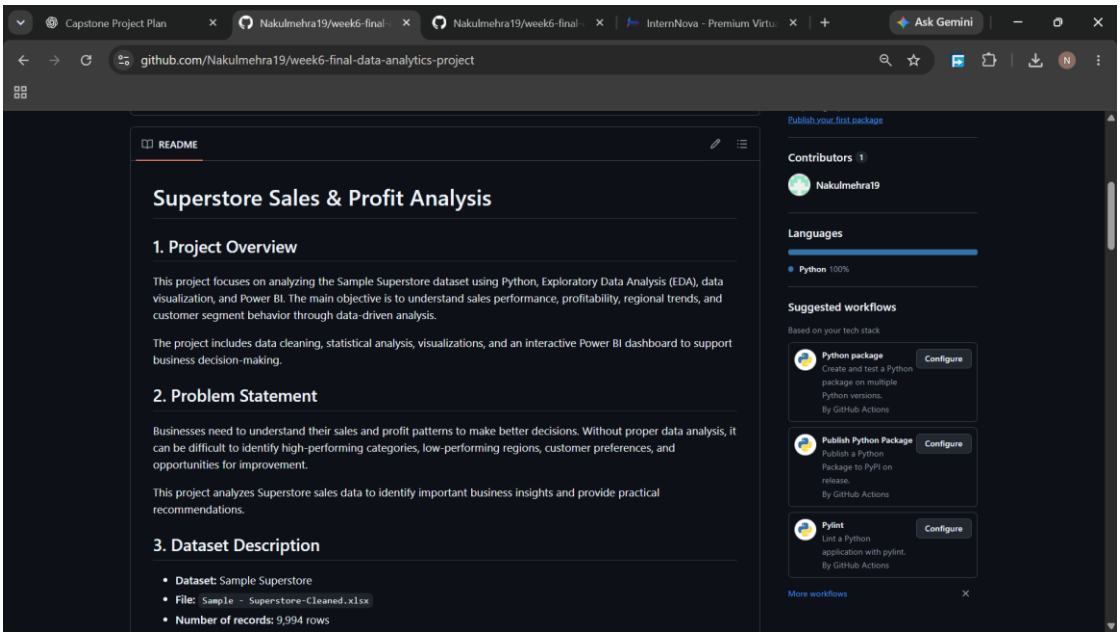


Figure 19. GitHub README showing project documentation.



Task 8 Screenshots — GitHub Documentation

Conclusion

This Week 6 project brings together Git/GitHub workflow, dataset preparation, exploratory analysis, Python visualization, Power BI dashboard development, business interpretation and professional documentation. The Sample Superstore analysis highlights differences by category, region and customer segment and provides a basis for investigating profitability and planning business activity.

Before final submission, verify the actual commit history, check the Order Count calculation, reset dashboard slicers, and insert the actual visualization/dashboard screenshots if the instructor expects images embedded in the report.

Appendix: Final Submission Checklist

- Confirm Git and GitHub workflow evidence is available if requested.
- Verify at least three meaningful commits in the GitHub commit history.
- Check that all Python scripts and eight PNG charts are in the repository.
- Check that the cleaned dataset path used by scripts is documented and available to the evaluator.
- Open the Power BI PBIX and test all slicers.
- Verify Order Count uses the intended aggregation.
- Confirm README and PowerPoint presentation are visible in GitHub.
- Insert actual screenshots into the DOCX if required by the assignment.
- Review the report and save a final copy before submission.

— END OF TASK-WISE REPORT —